

LAB EXERCISE - 10

Pathway Enrichment Analysis Using Enrichr

Question to be Solved:

Select genes that satisfy the following criteria:

Upregulated genes: Fold Change $\geq +2$

Downregulated genes: Fold Change ≤ -2

p-value ≤ 0.05

Perform pathway enrichment analysis on the filtered gene list using the Enrichr tool against:

(i) KEGG Pathway Database

(ii) Reactome Pathway Database.

Analyze the results and report the top significantly enriched pathways.

Workflow for Pathway Enrichment Analysis Using Enrichr:

1. Filter the genes based on given logFC and p-value. Copy the DEG's list.

ID	adj.P.Val	logFC	P.Value	t	B	Gene symbol	Gene title
56409_at	0.000119	1.1874491	4.15E-06	6.252785	10.43936	SORBS1	sortin and SH3 domain containing 1
56474_at	0.000119	1.0985451	4.54E-06	6.234937	10.35418	HSPB8	heat shock protein family B (small) member 8
45772_at	0.0001197	1.0185874	6.29E-06	6.170223	10.05261	SLMAP	sarcolemma associated protein
49932_t_at	0.000435	1.4310115	4.15E-06	5.787579	8.29891	MOC	mutated in colorectal cancers
43817_t_at	0.000493	1.1162385	6.29E-06	5.701851	7.91439	CLU	clusterin
46278_at	0.001182	1.50887	2.13E-07	5.440386	6.78349	MYLK	myosin light chain kinase
47481_at	0.001182	1.5629486	2.17E-07	5.440384	6.76858	ANGPTL1	angiotensin like 1
48039_at	0.001866	1.2553163	3.57E-07	5.333201	6.30701	SYNP2	synaptopodin 2
50361_at	0.002012	1.1653190	4.01E-07	5.307959	6.19944	SYNP2	synaptopodin 2
47941_at	0.002018	1.0240018	4.23E-07	5.298147	6.14622	CICR58	chromosome 8 open reading frame 58
52140_at	0.002052	1.2798888	4.67E-07	5.274548	6.05758	PGM5	phosphoglucomutase 5
42913_t_at	0.002052	1.0822762	5.20E-07	5.250962	5.9578	PLIN4	perilipin 4
45280_at	0.005578	1.2338624	2.22E-06	4.927108	4.61881	FAM107A	family with sequence similarity 107 member A
51939_at	0.006291	1.084874	2.94E-06	4.88316	4.3812	SORBS2	sortin and SH3 domain containing 2
56126_at	0.006891	1.3010469	4.80E-06	4.769375	3.90909	RPRM	reprimin, TP53 dependent G2 arrest mediator candidate
45320_at	0.009119	1.0893694	5.30E-06	4.728136	3.8177	DPH3	diphthamide biosynthesis 3
46399_at	0.0041063	1.0731981	4.12E-06	4.228135	1.9397	GATA5	GATA binding protein 5
51772_at	0.0056098	1.1241087	6.86E-06	4.097956	1.47517	CCBE1	collagen and calcium binding EGF domains 1
45794_at	0.0436689	1.2054271	1.78E-03	3.182024	-1.45533	KLHL41	ketch like family member 41
44856_r_at	0.0453741	1.0543639	1.94E-03	3.159606	-1.53058	VGLL3	vestigial like family member 3

2. Go to <https://maayanlab.cloud/Enrichr/> and Paste the DEG's list copied from the spreadsheet (each ID per line) in the Input Box given and Click Submit.

Enrichr

Login | Register

114,808,186 sets analyzed
465,796 terms
225 libraries

Analyze What's new? Libraries Gene search Term search About Help

Input data

Expand a gene, a term, or a variant into a gene set:
e.g. STAT3, breast cancer, or rs28897756

Drop a file or paste a set of Entrez gene symbols on each row in the textbox below. You can try a gene set example. Also, you can now try adding a background.

Try an example **STAT3 breast cancer rs28897756**

Include the top 100 most relevant genes

20 gene(s) entered

In order to enable others to search your set please enter a brief description of it:

☐ Contribute your set so it can be searched by others

Submit

3. After Submit, it loads and from the dashboard click Pathways

Enrichr

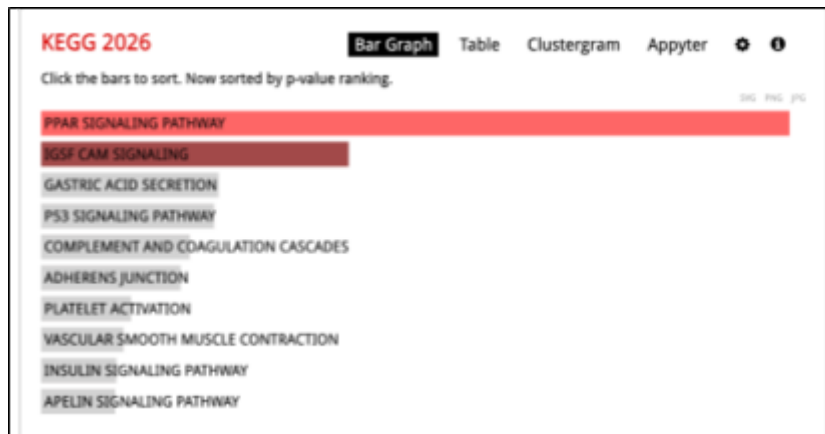
Login | Register

Transcription **Pathways** Ontologies Diseases/Drugs Cell Types Misc Legacy Crowd

Description: No description available (19 genes)

Reactome Pathways 2024 Smooth Muscle Contraction Smooth Muscle Contraction Smooth Muscle Contraction Smooth Muscle Contraction Smooth Muscle Contraction	WikiPathways 2024 Human Smooth Muscle Contraction Smooth Muscle Contraction Smooth Muscle Contraction Smooth Muscle Contraction Smooth Muscle Contraction	BioPlanet 2019 Smooth Muscle Contraction Smooth Muscle Contraction Smooth Muscle Contraction Smooth Muscle Contraction Smooth Muscle Contraction
WikiPathways 2024 Mouse Smooth Muscle Contraction Smooth Muscle Contraction Smooth Muscle Contraction Smooth Muscle Contraction Smooth Muscle Contraction	ARCH54 Kinases Coexp Smooth Muscle Contraction Smooth Muscle Contraction Smooth Muscle Contraction Smooth Muscle Contraction Smooth Muscle Contraction	KEGG 2026 Smooth Muscle Contraction Smooth Muscle Contraction Smooth Muscle Contraction Smooth Muscle Contraction Smooth Muscle Contraction
Elsevier Pathway Collection Smooth Muscle Contraction Smooth Muscle Contraction Smooth Muscle Contraction Smooth Muscle Contraction Smooth Muscle Contraction	MSigDB Hallmark 2020 Smooth Muscle Contraction Smooth Muscle Contraction Smooth Muscle Contraction Smooth Muscle Contraction Smooth Muscle Contraction	BioCarta 2016 Smooth Muscle Contraction Smooth Muscle Contraction Smooth Muscle Contraction Smooth Muscle Contraction Smooth Muscle Contraction
HumanCyc 2016 Smooth Muscle Contraction Smooth Muscle Contraction Smooth Muscle Contraction Smooth Muscle Contraction Smooth Muscle Contraction	NCI-Nature 2016 Smooth Muscle Contraction Smooth Muscle Contraction Smooth Muscle Contraction Smooth Muscle Contraction Smooth Muscle Contraction	Panther 2016 Smooth Muscle Contraction Smooth Muscle Contraction Smooth Muscle Contraction Smooth Muscle Contraction Smooth Muscle Contraction

4. Now, Reactome Pathways 2024 and KEGG 2026 from the list of Pathways is clicked



Pathway enrichment analysis using Enrichr revealed significant enrichment of metabolic, cytoskeletal, and immune-related pathways. **KEGG** analysis identified the **PPAR signaling pathway** as the top enriched pathway, followed by p53 signaling, complement and coagulation cascades, and adherens junctions.

Reactome analysis further highlighted **smooth muscle contraction**, RHO GTPase-mediated signaling, complement activation, and stress response pathways. Collectively, these results suggest that the filtered gene set is involved in metabolic regulation, cytoskeletal remodeling, immune activation, and cellular stress responses.